

# POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

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Enrollment No. N36245460

Jaypee Institute of Information Technology, Noida

T1 Examination, ODD 2024

B.Tech I<sup>st</sup> Semester

Course Title : Software Development Fundamentals – I Maximum Time : 1 hour

Course Code : 15B11CI111

Maximum Marks : 20

| Course Outcomes |                                                                                                                                                         | Cognitive Levels     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| CO1             | Explain the logic for solving problems considering various phases of Software Development Life Cycle and depicting them using algorithms and flowcharts | Understand (Level 2) |
| CO2             | Explain basics of C programming concepts to make decision for solving problems                                                                          | Understand (Level 2) |
| CO3             | Demonstrate and contrast different methods for writing modular programs in C                                                                            | Understand (Level 2) |
| CO4             | Use various C programming constructs to implement iteration, and recursion                                                                              | Apply (Level 3)      |
| CO5             | Apply and implement arrays, pointers structures and file handling for solving real-world problems                                                       | Apply (Level 3)      |

**Q1[CO1, (Understanding), 2+2 marks]** a) In the below table, there are two columns. You need to categorize the case studies given in 1<sup>st</sup> column into either of the SDLC model given in 2<sup>nd</sup> column. Briefly explain your reasons for the same.

| Case Study                                                                                                                                                    | SDLC model         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| i) A hospital is developing a comprehensive healthcare management system for patient records, appointment scheduling, billing, and medical inventory.         | a) Prototype model |
| ii) An educational institution wants to create a new e-learning platform with interactive features, including quizzes, video conferencing and student forums. | b) Waterfall model |

b) There are four persons A, B, C, D. Using boat, for crossing the river, A individually takes 1 minute, B takes 2 minutes, C takes 5 minutes and D takes 7 minutes. The boat can carry at most two people at a time, where the time taken in case of two people will be maximum time of both of them. Boat cannot come empty, someone has to row the boat. What is the minimum time in which all four can cross the river?

**Q2[CO1, (Understanding), 5 marks]** Design a flowchart to determine a student's eligibility for financial aid based on various criteria, including academic performance (GPA), family income, extracurricular involvement (Input: score between 1-10) and special circumstances (Input: Y for Yes and N for NO). *Continuing question in next page.*

**Eligibility criteria:**

- If the student's GPA is below 3.0, they are immediately disqualified from receiving aid, regardless of other factors.
- If the family income is low (below 30,000) the student qualifies for financial aid
- If the family income is medium (between 30,000 and 60,000), then extracurricular involvement score should be greater than 5 for providing aid.
- If the family income is high (above 60,000), the student qualifies only if they demonstrate extracurricular involvement (i.e. score > 5) and have special circumstances (Y)

*In Q3 below, you have been given two choices, you can attempt any one question of your choice.*

**Q3[CO2, (Understanding), 5 marks]** You are required to write a C program that simulates a simple ATM (Automated Teller Machine) system. The following features should be implemented in the program: i) Present the user with a menu of options: 1: Deposit Money 2: Withdraw Money 3: Check balance.

ii) When the user selects the "Deposit Money" option, ask them to enter the amount they wish to deposit. If the deposit amount is negative or zero, print an error message. If the deposit amount is valid (greater than zero), add it to the user's current balance and display a success message showing the updated balance.

iii) When the user selects the "Withdraw Money" option, ask them to enter the amount they wish to withdraw. If the withdrawal amount is negative, zero, or exceeds the current, print an error message (e.g. "Insufficient funds" or "Invalid amount"). If the withdrawal amount is valid (greater than zero and less than or equal to the current balance), subtract it from the user's balance and display a success message showing the updated balance.

iv) In case of "Check Balance", display the current balance amount.

**OR**

**Q3[CO2 (Understanding), 5 marks]** a) Write a C program using control flow statement (without using if-else) to give users the four choices to pick from the four bitwise operators you know and display the result after operating on user provided numbers in hexadecimal format. b) Solve and show the numerical steps to do the bitwise operations using the bitwise operators (AND, XOR) among these two numbers 0x3A924D7F and 0x4A2D5E8C.

**Q4** Please review the C code snippet below in i) and ii). Is there any possibility of an error? If yes, then how can it be fixed? If not, what will be the output?

|                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>i) [CO4 (Applying), 2 marks]</p> <pre>#include&lt;stdio.h&gt; int main( ) {     int i, j, k, n=4;     for(i=0; i&lt;n; i++){         for(j=0; j&lt;n; j++){             for(k=0; k&lt;=j; k++){                 printf("%d", k);             }             printf("\n");         }     }     return 0; }</pre> | <p>ii) [CO4 (Applying), 2 marks]</p> <pre>#include &lt;stdio.h&gt; int main( ) {     int i=1, j=1;     outer_loop: if(i&gt;3) { goto end;}     j=1;     inner_loop: if(j&gt;2){ j++; goto outer_loop;}     if(i==2 &amp;&amp; j==2) { goto skip_print;}     printf("i=%d, j=%d\n", i, j);     skip_print: i++;     goto inner_loop; end: printf("Loops terminated\n");     return 0; }</pre> | <p>iii) [CO2 (Understanding), 2 marks]</p> <p>Write the program in C for the program in ii) part to convert it into a program without goto, using any other control flow statement.</p> |
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